

Algebra II: Vectors and Parametric Equations

Formula and Concept Sheet

Vectors in Two and Three Dimensions

A vector records direction and length. Its components show the change in each coordinate:

$$\mathbf{v} = \langle a, b \rangle \quad \text{in 2D}, \quad \mathbf{v} = \langle a, b, c \rangle \quad \text{in 3D}.$$

If $P = (x_1, y_1)$ and $Q = (x_2, y_2)$, then

$$\overrightarrow{PQ} = \langle x_2 - x_1, y_2 - y_1 \rangle.$$

In 3D, include the change $z_2 - z_1$ as the third component.

For vectors with matching dimensions,

$$\begin{aligned}\langle a, b \rangle + \langle c, d \rangle &= \langle a + c, b + d \rangle, \\ k\langle a, b \rangle &= \langle ka, kb \rangle.\end{aligned}$$

The same rules apply component by component in 3D.

Length, Dot Product, and Angle

The length, or magnitude, of a vector is

$$\|\langle a, b \rangle\| = \sqrt{a^2 + b^2}, \quad \|\langle a, b, c \rangle\| = \sqrt{a^2 + b^2 + c^2}.$$

The dot product is

$$\langle a, b \rangle \cdot \langle c, d \rangle = ac + bd$$

and, in 3D,

$$\langle a, b, c \rangle \cdot \langle d, e, f \rangle = ad + be + cf.$$

For nonzero vectors $\mathbf{u}$ and $\mathbf{v}$, the angle ϕ between them satisfies

$$\cos \phi = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}, \quad 0 \leq \phi \leq \pi.$$

Useful tests:

$$\mathbf{u} \cdot \mathbf{v} = 0 \implies \mathbf{u} \perp \mathbf{v}, \quad \mathbf{u} = k\mathbf{v} \implies \mathbf{u} \parallel \mathbf{v}.$$

Components Parametrized by t

A vector-valued rule lets every component depend on one parameter:

$$\mathbf{r}(t) = \langle x(t), y(t) \rangle \quad \text{or} \quad \mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle.$$

Each value of t gives one point. The parameter need not represent time unless the context says it does.

A line through $P = (x_0, y_0)$ with direction vector $\langle a, b \rangle$ can be parametrized by

$$\mathbf{r}(t) = \langle x_0, y_0 \rangle + t\langle a, b \rangle,$$

or equivalently,

$$x = x_0 + at, \quad y = y_0 + bt.$$

To parametrize the line through points P and Q , use the direction vector $\overrightarrow{PQ}$. Different nonzero scalar multiples of a direction vector describe the same line.

In 3D, a line through (x_0, y_0, z_0) with direction $\langle a, b, c \rangle$ is

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct.$$

To remove a parameter, solve one component equation for t and substitute into the others.

Components Parametrized by θ

A circle of radius $R > 0$ centered at (h, k) has the standard parametrization

$$x = h + R \cos \theta, \quad y = k + R \sin \theta, \quad 0 \leq \theta < 2\pi.$$

As θ increases from 0, this starts at $(h + R, k)$ and travels counterclockwise. Every point satisfies

$$(x - h)^2 + (y - k)^2 = R^2$$

because $\cos^2 \theta + \sin^2 \theta = 1$. Swapping sine and cosine, inserting negative signs, or shifting θ can change the starting point or direction without changing the circle.

30-Minute Multiple-Choice Practice

Instructions. Complete all 12 questions in 30 minutes. Select the best answer for each question. A calculator may be used, but give exact values unless a decimal is requested.

1. If $\mathbf{u}(t) = \langle 2t - 1, t^2 - 4 \rangle$, what is $\|\mathbf{u}(3)\|$?
 - A. 5
 - B. $5\sqrt{2}$
 - C. 10
 - D. $10\sqrt{2}$
2. Which parametrization describes the line through $(-2, 5)$ and $(4, -1)$?
 - A. $x = -2 + t, y = 5 - t$
 - B. $x = -2 + t, y = 5 + t$
 - C. $x = 4 - 2t, y = -1 + 5t$
 - D. $x = -2 + 4t, y = 5 - t$
3. The line $x = 3 - 2t, y = -1 + 4t$ intersects the line $y = x$. What is the point of intersection?
 - A. $\left(\frac{1}{3}, \frac{1}{3}\right)$
 - B. $\left(\frac{5}{3}, \frac{5}{3}\right)$
 - C. $(1, 1)$
 - D. $(3, 3)$
4. Which Cartesian equation is traced by $x = 1 + 3t, y = -2 + 6t$?
 - A. $y = 2x - 4$
 - B. $y = 2x - 2$
 - C. $y = 3x - 5$
 - D. $y = 6x - 8$
5. Which parametrization traces the circle centered at $(-3, 2)$ with radius 5 counterclockwise, starting at its rightmost point when $\theta = 0$?
 - A. $x = -3 + 5 \cos \theta, y = 2 + 5 \sin \theta$
 - B. $x = -3 + 5 \sin \theta, y = 2 + 5 \cos \theta$
 - C. $x = 3 + 5 \cos \theta, y = -2 + 5 \sin \theta$
 - D. $x = -3 - 5 \cos \theta, y = 2 + 5 \sin \theta$
6. The equations $x = 4 + 3 \sin \theta, y = -1 - 3 \cos \theta$ trace which motion as θ increases from 0?
 - A. A radius-3 circle centered at $(4, -1)$, starting at the bottom and moving counterclockwise

- B. A radius-3 circle centered at $(4, -1)$, starting at the top and moving clockwise
- C. A radius-3 circle centered at $(-4, 1)$, starting at the bottom and moving counterclockwise
- D. A radius-9 circle centered at $(4, -1)$, starting at the right and moving clockwise

7. What is the angle between $\langle 3, 4 \rangle$ and $\langle -4, 3 \rangle$?
- A. 0°
 - B. 45°
 - C. 90°
 - D. 180°
8. What is the angle between $\langle 1, \sqrt{3} \rangle$ and $\langle 2, 0 \rangle$?
- A. 30°
 - B. 45°
 - C. 60°
 - D. 120°
9. For what value of k are $\langle k, 2 \rangle$ and $\langle 3, -6 \rangle$ perpendicular?
- A. -4
 - B. -2
 - C. 2
 - D. 4
10. What is the distance from $P = (1, -2, 3)$ to $Q = (5, 1, -3)$?
- A. $\sqrt{29}$
 - B. $\sqrt{52}$
 - C. $\sqrt{61}$
 - D. 13
11. What is the angle between $\langle 1, 2, 2 \rangle$ and $\langle 2, 1, -2 \rangle$?
- A. 30°
 - B. 60°
 - C. 90°
 - D. 120°
12. A line passes through $(2, -1, 4)$ and $(-1, 5, 1)$. At what point does the line intersect the xy -plane?
- A. $(-2, 7, 0)$
 - B. $(-1, 5, 0)$
 - C. $(0, 3, 0)$
 - D. $(2, -1, 0)$